

WEST BENGAL STATE ELECTRICITY DISTRIBUTION COMPANY LIMITED
(A Govt. of West Bengal Enterprise)

Office of the Chief Engineer,
Procurement & Contracts Department
VidyutBhavan, 4th. floor,
Bidhannagar, Kolkata – 700091.

Technical Specification

for

**1.1 KV Grade 1C x 630 SqmmXLPE
InsulatedPVC sheathed Armoured Aluminum
Cable**

TECHNICAL SPECIFICATION FOR 1.1 KV GRADE 1C X 630 SQMM XLPE INSULATED PVC SHEATHED ARMoured ALUMINUM CABLE

1.0 Scope :

This Specification covers the requirement of XLPE insulated PVC sheathed Cables for working voltages upto and including 1100 Volts suitable for LV side of Distribution Transformers, any other LV use in 3-Phase (Earthed) System.

2.0 Location :

2.1 The Cable shall be laid/buried directly in ground anywhere in West Bengal and terminated for outdoor connection.

2.2 The Cables may be laid within covered Cable Trenches in Cable Racks/Ladder/Air for certain portion of length.

3.0 Standards :

3.1 The Cables all conform broadly, to the latest issue of the following standards including all additional, alterations and modifications thereof.

3.2 IS : 7098 (Part-I) : 1988(With Latest amendment)
(Specification for XLPE insulated PVC sheathed heavy duty Electric Cables for working voltages upto and including 1100 Volts)

3.3 IS : 8130 : 1984, IS: 10462 (Part-I) : 1983, IS: 5831, IS: 7098 (Part-I) 1988, IS:10810-1984, IS:10418-1982,IS:3975-1979 with latest amendment.

4.0 Shape of Cables :

Cables shall be of circular shaped as per relevant IS.

5.0 Conductors :

5.1 Aluminium compacted Conductor complying with IS:8130:1984 shall be used (H4) Grade with flexibility of class-2.

5.2 The Conductors shall be stranded compacted aluminium conductor.

6.0 Insulation :

6.1 The Conductor shall be provided with XLPE insulation applied by extrusion. Thickness of insulation should be 2.8 mm (nom.) as per IS:7098 (Part-I): 1988 latest amendment. Thickness of insulation shall not fall below the nominal value(t_i) by more than $(0.1\text{mm} + 0.1 \times t_i)$.

7.0 Inner Sheath :

Inner sheath not applicable as per IS:7098 (Part-1).

8.0 ARMOURING :

Non-magnetic Aluminium rounded wire armouring shall be used. Hard drawn aluminium round wire of nominal diameter 2 mm with radial coverage above 90% shall be used as per the IS:8130:1984.

9.0 Outer Sheath :

The outer sheath shall be PVC ST-2 type. The colour of outer sheath shall be yellow. Thickness of outer sheath shall be 1.72 mm (min.) as per IS:7098 (Part I) – 1988 with latest amendment thereof.

10.0 Test and Test Certificates :

10.1 All the type tests and routine tests shall be carried out in the Manufacturer's Works as per IS:7098(Part-I) 1988 or latest amendment thereof as per IS-3975 & IS-5831.

10.2 The tests commensurate under 'Acceptance Tests' in the relevant IS shall be carried out as 'Acceptance Test'

10.3 All tests shall be performed in presence of Purchaser's representative, if so desired by the Purchaser. The manufacturer shall give at least thirty (30) days advance notice of shop tests.

10.4 All tests carried out at Works shall be furnished in six(6) copies for approval of the purchaser.

10.5 The Cable shall be dispatched from Works only after receipt of Purchaser's written approval of shop test reports.

11.0 **MANUFACTURER'S IDENTIFICATION :**

The Manufacturer's Name or Trade Mark shall be identified throughout the length of the Cable either by embossing on the outer sheath or by means of a tape bearing the following particulars.

- (a) Manufacturer's Name or Trade Mark
- (b) Voltage Grade, Type of cable, Type of insulation.
- (c) Nominal Section & material of Conductor and Number of Cores
- (d) Month and Year of Manufacture.
- (e) Inscription for length of Cables at 1.0 meters interval
- (f) Name of the Purchaser : WBSIEDCL
- (g) Marking "ELECTRIC" shall be embossed throughout the length of the Cable at 1.0 meter spacing (for XLPE Cable conforming to IS:7098 (Part-I) 1988 only).
- (h) Project name, if any.

12.0 **PACKING :**

- 12.1 The Cable shall be supplied in non-returnable wooden drums each containing length of 500 Meters of Cable with tolerance of $\pm 5\%$ per drum subject to overall maximum minus (-) 2% on total quantity will be accepted.

Cable Size	Drum Length
1C x 630 sq. mm	500 Mtr

- 12.2 The drums shall be proofed against attack by white ant or termite, conforming to IS:10418:1982

13.0 **MARKING :**

- 13.1 The following information shall be marked on each drum:

- (a) Drum identification No.
- (b) Manufacturer's Name, Brand Name or Trade Mark.
- (c) Nominal cross-sectional area of the Conductor of the Cable.
- (d) No. of Cores.
- (e) Type and Voltage of the Cable.
- (f) Length of the Cable on the drum.
- (g) Direction of rotation of Drum (by means of an arrow)
- (h) Approximate Weight : Tare : Gross
- (i) Year& Month of Manufacture.
- (j) Purchase Order No.
- (k) Name of the Purchaser : WBSIEDCL

- 13.2 The Drums may also be marked with ISI Certificate Mark, if applicable.

14.0 **DESCRIPTION :**

1.1 KV Grade single-core stranded Aluminium Conductor XLPE insulated, Non-magnetic roundarmoured, extruded PVC ST-2 outer sheathed conforming to IS:7098 (Part-I): 1988 read with its latest amendment and desired (as per tender specification) technical specification.

15.0 **DRAWING DATA & MANUAL :**

The following information shall be furnished in triplicate along with Tenders :

- (a) Manufacturer's leaflet giving construction details, dimensions and characteristics of different Cables.
- (b) Current rating of Cables including de-rating factor due to grouping, ambient temperature and type of various installations.
- (c) Write up, sketch illustrating the manufacturer's recommendation and splitting, jointing and termination of cables.
- (d) List of customers to whom the cable of similar rating have been supplied.

16.0 **CLASSIFICATION OF TESTS :**

- 16.1 **Type Test Report** Type Test Report for the type tests conducted in accordance with IS:7098(Part-I): 1988 within last 5 years from the due date of Opening of Tender on similar type of Cables (as per Tender Specification) in a NABL accredited Test House or Laboratory are to be submitted along with the tender otherwise tender may be rejected. Certificate should bear NABL Logo.
- a) Tests on conductor :ResistanceTest,TensileTest,Wrapping Test.
 - b) Tests for armouring Wires.
 - c) Test for thickness of insulation and sheath
 - d) Physical test for insulation.
 - i)Tensile strength and elongation at break.
 - ii)Ageing in air oven
 - iii) Hot set test
 - iv)Shrinkage test
 - v)Water absorption (Gravimetric)
 - e) Physical tests for outer sheath
 - i) Tensile strength and elongation at break
 - ii) Aeging of air oven
 - iii) Loss of mass in air oven
 - iv) Shrinkage tests
 - v) Hot deformation
 - vi) Heat shock test
 - vii) Thermal stability
 - f)Insulation resistance (VOLUME RESISTIVITY) TEST
 - g) High voltage Test
 - h) Flammability Test
- 16.2 **Acceptance Test:** The following shall constitute Acceptance Tests :
- a) Tensile Test
 - b) Wrapping Test
 - c) Conductor resistance test.
 - d) Test for thickness of insulation and sheath.
 - e) Hot set test for insulation.
 - f) Tensile strength and elongation at break test for insulation and outer sheath.
 - g) High Voltage test.
 - h) Insulation resistance (VOLUME RESISTIVITY) TEST
- 16.3 **Routine Test:** The routine test shall be carried out on all cables manufactured in accordance with this specification. The following routine tests shall be made on cable length as specified in the ISS.
- a) Conductor resistance test.
 - b) High voltage test.
- 17.0 Valid Calibration Certificate of instruments/equipment used for testing purpose conducted by NABL accredited Laboratory provided the certificate bears an accreditation body logo. Documents to be submitted at the time of physical delivery at consigned stores.
- 18.0 **Documents to be submitted at the time of physical delivery at consignee stores:**
- The following documents are to be submitted by the venders to the consignee stores at the time ofDispatch to stores by the venders:
- a) Copy of Purchase Order
 - b) Copy of dispatch instruction
 - c) Inspection Test certificate
 - d) Guarantee certificate
 - e) Proforma Invoice
 - f) Calculation Sheet for price variation on the basis of IEEMA or CACMAI as applicable with base date of order
 - g) Seal list and packing list
 - h) Challan in triplicate
 - i) Way Bill, if applicable

GUARANTEED TECHNICAL PARTICULARS FOR		
1.1 KV Grade 1C x 630 Sqmm XLPE Armoured Aluminum Cable		
SL. NO	NAME OF MANUFACTURER	M/s.-----
A)	Size of Cable	1C x 630 sq. mm
	Rated Voltage	1100V
	Cable Code	A2XWαY
B)	Standard Referred	IS: 7098(Part-I), IS: 1554(1), 8130, 5831, 10810, 10462 & 3975 with latest ammendment
C)	Conductor	
	i) Material	H4 Grade Al of Class II as per IS:8130/84
	ii) Nominal Area of Cross-section (sq. mm)	630
	iv) Shape of conductor	Stranded Circular
	v) Min. No of Strand	53
D)	Insulation	
	i) Material	XLPE as per IS:7098 (Part-I):1988
	ii) Nom. Thickness (mm)	2.8
	iii) App. Dielectric strength (Kv/mm)	25-30 KV/mm
	iv) Suitability with regard to temperature, moisture, acid, oil and alkaline surrounding	Yes
E)	i) Maximum conductor temperature (Deg.)	90
	ii) Maximum conductor temperature during short ckt. (Deg.)	250
	iii) Drum length	500 mtr.
	iv) Tolerance of overall quantity	-2%
F)	Armouring	
	a) Material & Type	Hard drawn Aluminium round wire
	b) Nom.dia/dimen of armour wires/Strips in mm	2.0
G)	Outer sheath	
	i) Material	PVC Type -ST-2 as per IS:5831
	ii) Minimum thickness of sheath (mm)	1.72
	iii) Colour	Yellow
H)	Approx. overall dia of Cable (in mm)	
I)	Embossing	As per Clause 13.1
J)	Electrical Properties :-	
	i) Maximum d.c. resistance of conductor at 20 ⁰ C (Ohm/Km)	0.0469
	ii) Maximum permissible conductor temperature(°C) under full load	90°C
	iii) Rated volatge	1100V
	iv) Maximum operating voltage	1100V
	v) Permissible voltage variation	±10%
	vi) Rated frequency	50HZ
	vii) Permitted frequency variation	±5%
	viii) Minimum specific insulation resistance at 90 ⁰ C (Ohm-Cm)	1 x 10 ¹² ohm.cm
	ix) Continuous current carrying capacities As per 3961 (Part - VI) 2016	
	In Air at 40 ⁰ C (Amps)	793
	In Duct	477
	In Ground at 30 ⁰ C (Amps)	546

**Signature with Designation & Seal
With Name of the Firm**